

**Nguyen Khac Bao**

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May 11, 2026

**Hiring Team**

Nexlab Technology

Ho Chi Minh City, Vietnam

**Re: Application for Intern AI/ML/Data Engineer**

Dear Hiring Team at Nexlab Technology,

I am writing to apply for the Intern AI/ML/Data Engineer position. As an Information Technology student with a strong interest in RAG systems, LLM integration, and data pipelines, I am interested in Nexlab's work around AI agents for data analysis, automated reporting, and the modern data stack on Google Cloud.

Instead of only experimenting with models in notebooks, I focus on building small but complete AI systems. Recently, I developed a Vietnamese Legal RAG Chatbot, where I built a FastAPI backend, integrated Qdrant vector retrieval, designed an agent-style RAG workflow with LlamaIndex ReAct tools, and connected Gemini/OpenAI-compatible endpoints for chatbot responses. I also fine-tuned Llama-3.1-8B with PEFT/LoRA on Vietnamese legal QA data using Vast.ai GPU.

Beyond the chatbot, my FallGuard and Traffic Violation projects helped me practice data engineering fundamentals: raw data ingestion, CSV/JSON metadata tracking, data quality checks, annotation review, dataset splitting, and reproducible export workflows. These experiences align with the role's focus on clean data flow for AI models and dashboards.

I understand that Nexlab's stack includes tools such as BigQuery, Cloud Storage, Vertex AI, Looker, and Power BI. While my current hands-on cloud experience is still basic, mainly with Google Cloud Run microservices, AWS EC2, Docker, and Vast.ai GPU, I am comfortable self-learning from technical documentation and applying new tools in real projects.

I am available for a full-time internship and would be glad to contribute to RAG chatbot development, ETL/ELT pipelines, SQL-based data processing, and AI reporting workflows at Nexlab. Thank you for your time and consideration.

Sincerely,

**Nguyen Khac Bao**